# Dr.-Ing. Yinan Xiao

ynxiao@princeton.edu
www.linkedin.com/in/y1nan-x1ao
orcid.org/0009-0008-4606-5201
github.com/Yinan-Xiao

I am a Postdoctoral Research Associate at Princeton University, holding a Dr.-Ing. (summa cum laude) from Technische Universität Braunschweig, Germany, with doctoral co-supervision at the Technical University of Munich. My research focuses on physics-grounded intelligent structural design, aiming to develop the next generation of computational methods and tools that transform how structures are conceived, explored, and realized. By integrating structural mechanics, geometric computation, and artificial intelligence (AI), I seek to enable designers to interactively generate, evaluate, and refine structurally informed and fabrication-aware solutions. My long-term vision is to establish a new design paradigm that bridges architectural creativity, structural intelligence, and digital fabrication. (Portfolio link)

## Appointments

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| 10.2026 – | Postdoctoral Researcher from Adel Research Group (ARG) at School of Architecture, **Princeton University** (Advisor: Prof. Dr. Arash Adel) |
| 01.2026 – 05.2026 | Post-doc Researcher in StructurAI Lab in Department of Architecture, **The University of Hong Kong** (Advisor: Prof. Dr. Kam-Ming Mark Tam) |
| 10.2025 – 01.2026 | Research Assistant in [S]Equilibrium Lab in School of Architecture, **The Chinese University of Hong Kong** (Advisor: Prof. Dr. Shuaizhong Wang) |
| 10.2023 – 05.2024 | Scientific Researcher, Professorship of Structural Design in Architecture Department, **Technical University of Munich** (Advisor: Prof. Dr. Pierluigi D'Acunto) |
| 02.2021 – 09.2023 | Project Researcher, Collaborative Research Centre AMC TRR 277, **Technische Universität Braunschweig** (Advisor: Prof. Dr. Norman Hack & Prof. Dr. Harald Kloft) and **Technical University of Munich** (Advisor: Prof. Dr. Pierluigi D'Acunto) |
| 02.2021 – 09.2023 | Scientific Researcher, Digital Building Fabrication Laboratory (DBFL) in Faculty of Architecture, Civil Engineering and Environmental Sciences, **Technische Universität Braunschweig** (Advisor: Prof. Dr. Norman Hack & Prof. Dr. Harald Kloft) |
| 12.2019 – 09.2020 | Research Assistant, Design of Structure (ADS) in Department of Architecture, **Aalto University** (Advisor: Prof. Dr. Toni Kotnik) |

## Education

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| 02.2021 – 09.2025 | **Doctor of Engineering (Dr.-Ing.)** in Architecture, Civil Engineering and Environmental Sciences, Technische Universität Braunschweig, Germany,<br>graduating **Summa Cum Laude** with the highest academic distinction |
| 10.2023 – 05.2024 | Doctoral Research Co-supervised by Professorship of Structural Design in Architecture Department, Technical University of Munich, Germany |
| 09.2016 – 05.2020 | **Master of Science (M.Sc.)** in School of Structural Design and Architecture, Aalto University, Finland |
| 09.2017 – 01.2018 | Erasmus Exchange Study in Department of Architecture and Civil Engineering, University of Bath, United Kingdom |
| 09.2011 – 06.2016 | **Bachelor of Architecture (B.Arch.)** in Architecture Department, Huazhong University of Science and Technology, China |

## Grant Proposal Experiences

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| 06.2026 – 06.2028 | **Independent applicant and lead proposal author**, Learning Maxwell Reciprocity: A Graph Neural Framework for Continuous Shell Geometry and Force Duality, *Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) - Walter Benjamin Fellowship*; Host: Sigrid Adriaenssens, unfunded |
| 06.2025 – 12.2026 | **Main participant**, Upcycling Construction Waste: A Circular Construction Approach Through Automation and Digital Fabrication, *IdeaBooster Fund (IBF), The Chinese University of Hong Kong*; PIs: Shuaizhong Wang, funded HK$90,000 |
| 04.2026 – 03.2027 | **Main participant**, MiC from Waste: Integrating Concrete Component Reuse into Hong Kong's MiC System, *Knowledge Transfer Impact Project Fund (KT-IPF)*; PIs: Shuaizhong Wang, funded HK$200,000 |
| 01.2024 – 12.2027 | **Contributed to proposal development and writing**, Injection 3D Concrete Printing (I3DCP) - Material Efficient Lightweight Reinforced Concrete Structures based on Spatially Complex Strut-and-Tie Models, *Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) within the Collaborative Research Centre (CRC) / Transregio 277 – "Additive Manufacturing in Construction (AMC)" (Project A09), grant number TRR 277/1 2020 – Project number 414265976*; PIs: Norman Hack, Inka Mai, funded €700,000 (subproject budget) |
| 01.2020 – 12.2023 | **Contributed to proposal development and writing**, I3DCP – Injection 3D Concrete Printing, *Volkswagen-Stiftung (Volkswagen Foundation)*; PIs: Norman Hack, funded €300,000 |

## Awards & Honors

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| 05.2023 – 11.2023 | Time Space Existence: Venice Architecture Biennale (Venice, Italy) [Video] |
| 09.2022 | Best Presentation Award in Digital Concrete Conference (Loughborough, United Kingdom) |

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| 08.2020 – 08.2024 | Research Scholarship at Technische Universität Braunschweig supported by the Gerhard and Karin Matthäi Foundation (Braunschweig, Germany) |

## Additional Training & Certificates

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| 10.2023 – 03.2024 | Structural Optimization 1, by Chair of Statics and Dynamics (LSD, Prof. Dr.-Ing. Kai-Uwe Bletzinger), **Technical University of Munich** (Munich, Germany) |
| 10.2022 – 03.2023 | Robotics 1: Technical and Mathematical Foundations, by Institute for Robotics and Process Control (IRP, Porf. Dr. Jochen Jakob Steil), **Technische Universität Braunschweig** (Braunschweig, Germany) |
| 01.2021 – 05.2021 | 6.86x: Machine Learning with Python-From Linear Models to Deep Learning, by **MITx** (Online) |

## Teaching Experiences

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| 12.2025 – 04.2026 | Tutor & Coding tutorials, Building Technology 1 - Building Principles (Course Code: ARCH2056_ARCH7330) in **The University of Hong Kong, China** |
| 06.2025 – 07.2025 | Instructor, 15th Digital Futures summer workshops in **Tongji University, China** |
| 10.2022 | Guest lecturer, An Introduction of AI-Assisted Urban Design Methods (Course Code: A010125) in **Xi'an University of Architecture and Technology, China** |
| 08.2021 – 11.2021 | Instructor, Digital Building Fabrication Design Studio in **TU Braunschweig, Germany** |
| 01.2021 – 04.2021 | Instructor, LafargeHolcim 3D Printing Housing Hack (semester course). The result demonstrators won Holcim 3D Printing Hackathon competition 2nd and 4th prizes in **TU Braunschweig, Germany**. |

## Publications

† → Corresponding author

### Journal Articles

J1. Shen†, Y. & **Xiao†, Y.** Alternating Direction Residual Force Reduction. *Computer Methods in Applied Mechanics and Engineering,* 119021. ISSN: 0045-7825. https://doi.org/10.1016/j.cma.2026.119021 (2026).

J2. **Xiao, Y.**, Vandenberg, A., Lowke, D., Mai, I., D'Acunto, P., Kloft, H. & Hack, N. Automated Robotic Assembly Planning of Space Trusses for Injection 3D Concrete Printing. *Automation in Construction,* 106440. ISSN: 0926-5805. https://doi.org/10.1016/j.autcon.2025.106440 (2025).

J3. **Xiao, Y.**, Hack, N., Kloft, H., Lowke, D., Mai, I. & D'Acunto, P. Constraint-Based Form-Finding of Space Trusses for Injection 3D Concrete Printing Through Vector-Based Graphic Statics. *Additive Manufacturing,* 104751. ISSN: 2214-8604. https://doi.org/10.1016/j.addma.2025.104751 (2025).

J4. Hack, N., Kloft, H., Mai, I., D'Acunto, P., **Xiao, Y.** & Lowke, D. Injection 3D Concrete Printing: From Structural Geometry to Fabrication. *Architectural Design* **94.** https://doi.org/10.1002/ad.3094 (4-2024).

### Peer-reviewed Conference Proceedings

C1. Hack†, N., **Xiao†, Y.**, Jantzen, C., Tesche, J. & Mawas, K. *Design and Fabrication of Structures with Graphic Statics and Augmented Reality* in *Proceedings of IASS Annual Symposia* [URL] (ETH Zurich, 2024).

C2. He, F., **Xiao, Y.**, Shen, Y., Asadi, H., Wüchner, R. & D'Acunto, P. *Enhancing Non-Linear Force Density Method through Combinatorial Equilibrium Modelling* in *Proceedings of IASS Annual Symposia* [URL] (ETH Zurich, 2024).

C3. Shen†, Y., **Xiao†, Y.**, He, F. & D'Acunto, P. *Structural Form-finding Integrating Vector-based Graphic Statics and Non-Linear Force Density Method* in *Proceedings of IASS Annual Symposia* [URL] (ETH Zurich, 2024).

C4. **Xiao, Y.**, Hack, N. & Kloft, H. *Digital Structural Design And Shotcrete 3D Printing Strategies for Lightweight Reinforced Concrete Beam-Grid Structures* in *Proceedings for the 6th fib International Congress* (Oslo, 2022).

C5. **Xiao, Y.**, Khader, N., Vandenberg, A., Lowke, D., Kloft, H. & Hack, N. *Injection 3D Concrete Printing (I3DCP) Combined with Vector-Based 3D Graphic Statics* in *Third RILEM International Conference on Concrete and Digital Fabrication. DC 2022. RILEM Bookseries* **37** (Springer International Publishing, Cham, 2022), 43–49. ISBN: 978-3-031-06116-5. https://doi.org/10.1007/978-3-031-06116-5_7.
– Best Paper Presentation Award (Conference Digital Concrete 2022).

C6. **Xiao, Y.**, Shen, Y., Hack, N. & D'Acunto, P. *A structure-and fabrication-informed strategy for the design of lattice structures with Injection 3D Concrete Printing* in *Proceedings of IASS Annual Symposia* [URL] (Beijing, 2022), 1–12.

## Tools & Software

ForceDensityMethod: A form-finding toolkit featuring a collection of published extended functions based on the force density method.

I3DCP: Automatic robotic motion planner for Injection 3D Concrete Printing (I3DCP) technique. Detailed at [J1].

ADRFR: Alternating direction residual force reduction (ADRFR) method for form finding. Equal contribution with Y. Shen†.

## Academic Advising

### Graduate

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| 2023 | Changyang Chao, Leibniz University Hannover, Germany |
| 2022 | Xinjie Wang, TU Braunschweig, Germany |

## Academic Service

### Journal Reviewer

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| 2023 | Automation in Construction |

## Other Experience

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| 02.2018 – 08.2018 | Trainee at KSP Jürgen Engel Architekten, Frankfurt am Main, Germany |
| 05.2017 – 01.2020 | Instructor (remotely), Studio Alpha, Shanghai, China |

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Last updated: November 1, 2026